

What generalises from a frozen structure-model trunk is the local term, not the pooled one

Experiment 14 · ddG_with_Boltz · frozen Boltz-2 pair track · MLP · two blind corpora

A $\Delta\Delta G$ regressor on a frozen structure-prediction trunk must reduce the pair representation at a mutated residue to a fixed-length vector. We tested three biology-motivated additions to that readout — contact-weighted pooling, explicit burial and biophysics, and explicit evolutionary conservation. **All three fail.** The experiment's controls instead show that the **uniform mean over the whole chain behaves like a far-shell readout**, and that the single most local feature available — the diagonal element $z[i,i]$ — matches every 256-dimensional construction on blind transfer at **half the width**, while beating a plain substitution lookup by **+0.26 Pearson on both corpora**. The mechanism is a train/test mismatch: whole-chain pooled *levels* carry corpus-specific protein context that damages transfer, though it demonstrably helps in-distribution.

1. Motivation

The standard readout for a mutated residue i is the mean of the pair-track row $z[i, :]$ over all residues of the chain. In a 60-residue domain that dilutes the ~ 10 residues actually in contact by a factor of six; in a 300-residue protein, by thirty. The choice is widespread — the closest published method built on AlphaFold2 pair representations enumerates four aggregations (global mean or sum, mutation-site mean or sum), none spatially weighted, and Boltz-2's own affinity module "performs mean pooling over all pairwise interactions". Since folding stability at a residue is physically a property of the residues it packs against, we asked whether adding biology to this readout helps, and used matched-dimension controls to establish what the readout is actually doing.

2. Methods

Constructions. From the pair row at the mutated site we build: the **diagonal** $z[i,i]$ (unpooled); the uniformly **pooled difference** between mutant and wild type; the uniformly **pooled levels** (wild-type and mutant vectors side by side); and **contact-weighted** versions of the pooled quantities, where the weight is $P(d_{ij} < 8 \text{ \AA})$ read from the model's own predicted distogram at the wild-type structure, masked to $|i-j| > 2$. Two matched-dimension controls: a **far-shell** pooling using the renormalised complement $P(d \geq 8 \text{ \AA})$, and a **substitution-identity** baseline of 40 one-hot dimensions with no structural content. A biophysical block (40 d: burial, volume, hydropathy, transfer free energy, charge, secondary-structure propensity, Gly/Pro flags, and their interactions with burial) and a conservation block (14 d: alignment depth and effective depth, column entropy, PSSM log-odds, consensus indicator) were tested as additions.

Evaluation. Training is 12,359 mutations over 412 proteins with a five-fold protein-grouped split and a five-seed MLP ensemble; the fold models are averaged onto two blind corpora, S669 (541 variants / 62 proteins) and FireProt ≤ 500 (3,205 / 138). Because mutations within a protein share a structure, an embedding and an assay, all intervals come from a **cluster bootstrap over proteins** (400 resamples), reported as the paired difference against a reference construction on the same resample. Antisymmetry augmentation (adding each reverse mutation with negated $\Delta\Delta G$ to the training folds) is treated as an experimental factor rather than a fixed default; transfer numbers below are without it. FireProt is filtered to the 130 proteins sharing no 30 %-identity cluster with the training corpus; S669 contains no such homologues.

3. Results

3.1 A ladder of readouts

construction	dims	S669	FireProt ≤ 500	in-distribution
substitution identity (one-hot)	40	0.295	0.381	0.505
far-shell pooling	256	0.345	0.504	0.740
uniform pooling, levels	256	0.381	0.476	0.801
contact-weighted levels	256	0.487	0.657	0.793
diagonal + uniform pooled difference	256	0.541	0.647	0.792
diagonal + contact-weighted difference	256	0.547	0.665	0.791
diagonal alone	128	0.557	0.647	0.775

Table 1 — Blind-transfer Pearson r by construction. Same trunk, same model, same data; only the readout differs. The rightmost column is the in-distribution protein-holdout score on the training corpus.

The ordering is identical on two independently curated corpora with different label provenance. Uniform pooling of levels — the most common choice, and the best of these readouts *in-distribution* — sits third from the bottom on both blind sets, closer to far-shell pooling and to an amino-acid lookup than to the diagonal. Replacing it with the diagonal alone halves the readout and gains, on S669, Δr **+0.173 [+0.055, +0.285]**.

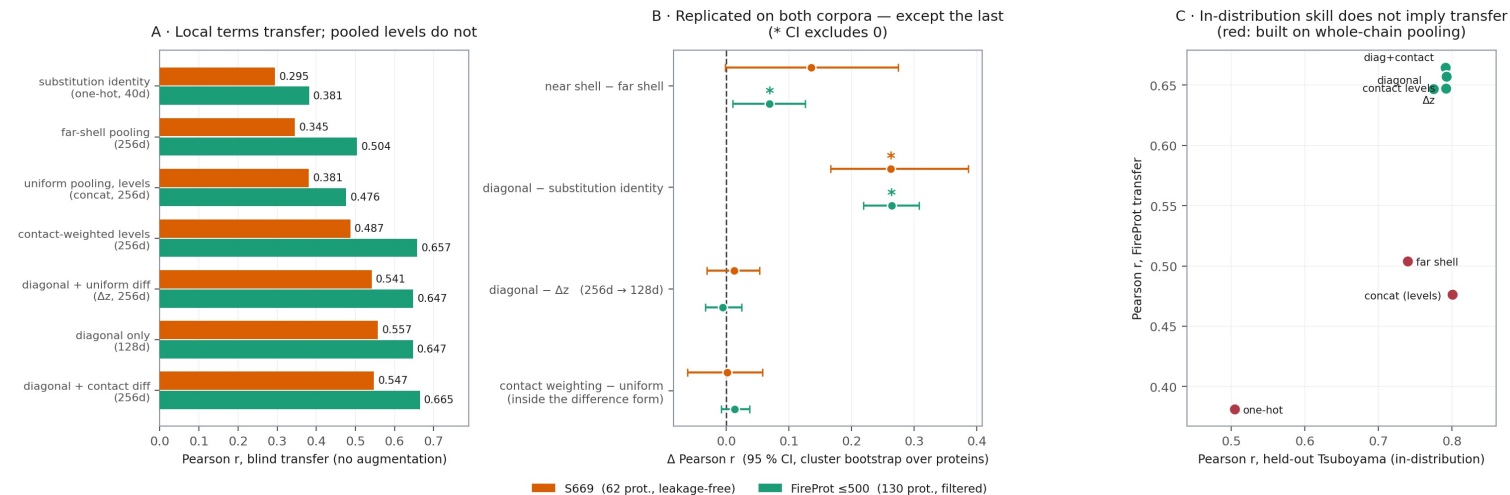


Figure 1. **A** The ladder of readouts on both blind corpora. **B** Paired bootstrap differences; the first three claims replicate on both corpora, the fourth does not. **C** In-distribution skill plotted against transfer skill: the configurations built on whole-chain pooling (red) are the strongest in-distribution and the weakest on transfer.

3.2 The local term is not a substitution lookup

Because $z[i, i]$ is the mutated residue's own pair element, the deflationary reading is that it encodes which amino acid changed into which. It does not: against 40 one-hot dimensions the diagonal gains **+0.263 [+0.167, +0.386]** on S669 and **+0.264 [+0.219, +0.308]** on FireProt — the same effect size, twice.

3.3 Uniform pooling is a far-shell readout; levels are the problem, differences are not

A matched-dimension far-shell control scores 0.345 / 0.504, statistically indistinguishable from uniform pooling of levels — as expected, since a uniform mean over a chain is dominated by distant residues. Against it, local constructions gain **+0.201 [+0.087, +0.315]** (S669) and **+0.069 [+0.010, +0.126]** (FireProt).

The decisive distinction is between pooled *levels* and pooled *differences*. The difference cancels the per-protein offset and is neutral: dropping it entirely (diagonal versus diagonal-plus-difference) changes nothing on transfer, +0.012 [-0.031, +0.053] and -0.006 [-0.034, +0.024]. The levels retain that offset, and it is corpus-specific: substituting them for the difference costs **-0.141 [-0.241, -0.020]** on FireProt.

3.4 The three biology-motivated additions

addition	comparison	S669	FireProt ≤500
contact weighting	Δ Pearson r	+0.001 [-0.062, +0.058]	+0.013 [-0.008, +0.037]
	Δ Spearman ρ	+0.007 [-0.058, +0.056]	+0.024 [+0.001, +0.045]
burial + biophysics	r, alone / added	0.014 alone; added to the baseline it does not exceed it	
MSA conservation	Δ Pearson r	—	+0.001 [-0.014, +0.019]
	Δ bias, stabilizing	—	+0.137 [+0.079, +0.182]

Table 2 — Paired differences for each addition. Bold marks an interval excluding zero.

Contact weighting does not replicate. On FireProt it reaches significance on rank correlation and error but never on Pearson r; on S669 the Pearson effect is exactly zero. **Biophysics** never exceeds the baseline, and its raw chain-scale features actively degrade transfer by importing more corpus-specific context. **Conservation** adds nothing over the pair track on a corpus with 100 % alignment coverage and a median alignment depth of 9,474 — consistent with the trunk having been given the alignment already.

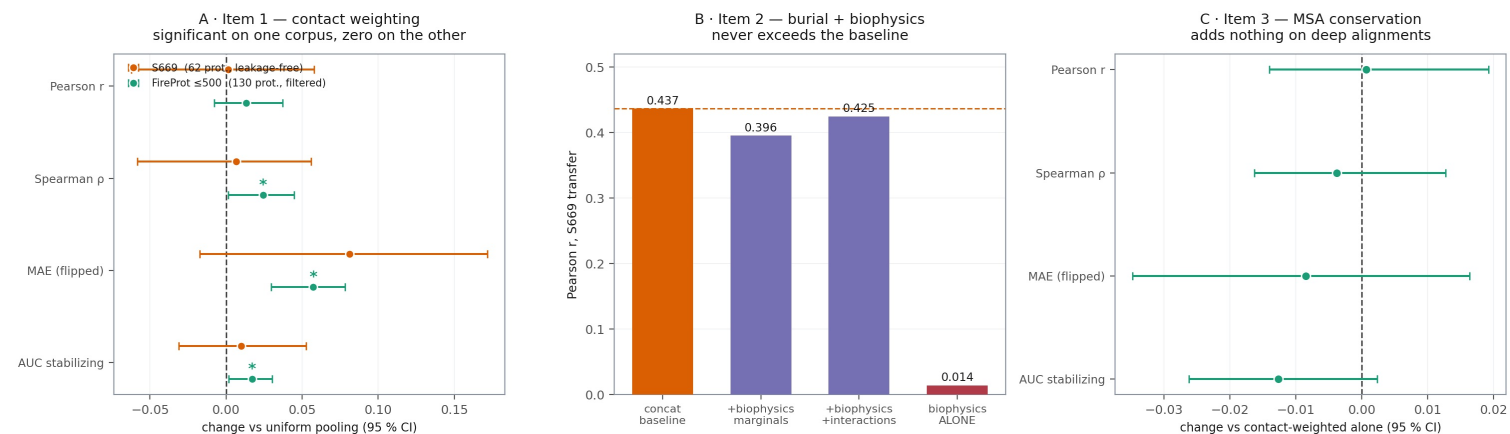


Figure 2. The three additions. **A** Contact weighting: significant on one corpus, zero on the other. **B** Biophysics never exceeds the baseline. **C** Conservation adds nothing over contact-weighted features on deep alignments.

4. Interpretation

Three additions that look like domain knowledge — spatial weighting, explicit biophysics, explicit conservation — add nothing to a trunk already trained on structures and alignments. What the trunk lacks is not information but an appropriate readout: the transferable signal sits in the local pair element, and the conventional whole-chain average both dilutes it and imports a per-protein offset that does not survive a change of corpus.

The mechanism is a train/test mismatch rather than a defect of the pooled features. In-distribution the pooled half is worth **-0.017 [-0.023, -0.010]** — it genuinely helps when train and test share a corpus. This makes in-distribution holdout performance an actively misleading model-selection signal here: the readout that ranks first in-distribution ranks near-last on both blind corpora.

5. Limitations

Neither blind corpus is the published benchmark. Both are capped at 500 residues: S669 here is 541 of 669 variants (62 of 94 proteins), and FireProt is likewise length-capped. Every construction is scored on the identical subset, so the comparisons above are valid and the cap cancels — but the absolute values are **not comparable to published figures** for these benchmarks, and the cap plausibly removes the hardest cases, since the training corpus consists of 32–72 residue domains. **No claim of state-of-the-art performance is made.**

Two corpora are not many. The ordering in Table 1 is identical on both, and the two largest effects replicate with near-identical point estimates, but both blind sets are literature-derived stability data; a third corpus with a different assay would strengthen the claim, particularly for the smaller effects.

- The backbone is frozen throughout. Conclusions about redundancy apply to a fixed representation and need not hold if the trunk were fine-tuned — in particular, a spatial weighting that adds nothing to frozen features might matter to a trainable readout.
- Conservation was tested only as *appended per-column statistics*. Whether altering what the trunk itself sees (mutating the alignment rather than only the query) behaves differently is untested and requires re-running the structure model.
- Ranking *within* the stabilizing tail is unchanged by every construction here, as it was by earlier loss-reweighting work in this project. That deficit remains open.
- Top-k detection precision proved unusable at this sample size, returning opposite signs on different corpora; no claim rests on it.

6. Conclusion

Asked what biology to add to a $\Delta\Delta G$ readout built on a frozen structure-prediction trunk, the empirical answer is: none of the obvious candidates, and the productive change is a deletion. The diagonal pair element alone, at half the dimensionality of the current representation, matches or exceeds every richer construction on blind transfer and improves substantially on the conventional uniform-mean readout. Where a strong pretrained trunk is used frozen, the readout deserves auditing — against matched-dimension controls, and on a corpus other than the one used to select it — before the feature list is extended.